

Dear Editors,

Please consider this Research Article, “Source-only model selection and limits of fixed spectral-sharing pipelines in cross-subject motor-imagery EEG decoding”, for publication in PLOS ONE.

The study examines how source-only training-duration selection and participant heterogeneity constrain claims about calibration-free motor-imagery EEG decoding. It combines a model-selection audit with frozen external evaluation and retains adverse and uncertain findings. The contribution is a reproducible selection and evaluation evidence chain, rather than a new state-of-the-art domain-generalization algorithm. A separate sensor-level physiological analysis is descriptive and does not establish a decoder mechanism.

The manuscript provides documented methods, participant-level outcomes, uncertainty estimates and explicit limitations. Code, manuscript candidates and frozen research outputs are publicly accessible on GitHub. The author reports no prior journal or preprint-platform publication.

Original EEG is obtained from the original providers; saved predictions, derived summaries and validation evidence are supplied or linked. AI assistance and author verification are disclosed in the Methods. Funding and competing-interest statements are provided in the submission fields.

I am the sole and corresponding author and have approved the manuscript. I take responsibility for its scientific content. I have no opposed reviewers. Suggested Academic Editors are Marie-Constance Corsi (Inria Centre de Recherche de Paris) and Cota Navin Gupta (Indian Institute of Technology Guwahati), whose expertise covers EEG motor imagery, decoding and signal analysis.

Sincerely,

Ziyuan Zhu

College of Artificial Intelligence Medicine, Chongqing Medical University, Chongqing, China
zzy2630816871@gmail.com

ORCID: <https://orcid.org/0009-0005-1153-4926>